Calculation Roughbook

Step 1: Measures of Central Tendency & Dispersion

- **Math_Score Statistics:**
  - Mean (μ): $\frac{\sum x}{N} = 70.02$
  - Median: Middle value of sorted scores = 70.30
  - Mode: Most frequent score = 100.0
- **Science_Score Statistics:**
  - Range: $\text{Max} - \text{Min} = 100.0 - 15.2 = 84.80$
  - Sample Variance (s^2): $\frac{\sum(x_i - \bar{x})^2}{n - 1} = 191.93$
  - Sample Standard Deviation (s): $\sqrt{191.93} = 13.85$

Step 2: Probability Basics

- **Probability of Passing:**
  - $P(\text{Pass}) = \frac{\text{Total Passes}}{\text{Total Students}} = \frac{3032}{5000} = 0.6064\ (60.64\ \%)$
- **Contingency Table (**Pass_Fail **vs** $\text{Hours_Studied} > 5$**):**

Pass Fail	False (≤5 hrs)	True (>5 hrs)	Total
Fail	1,960	8	1,968
Pass	34	2,998	3,032
All	1,994	3,006	5,000

- **Conditional Probability:**
  - $P(\text{Pass} \mid \text{Hours} > 5) = \frac{P(\text{Pass} \cap \text{Hours} > 5)}{P(\text{Hours} > 5)} = \frac{2998 / 5000}{3006 / 5000} = \frac{2998}{3006} = 0.9973\ (99.73\ \%)$

Step 3: Distribution & Visualization Metrics

- **Science_Score Distribution Shape:**
  - Skewness: -0.1020 (Slightly left-skewed / negative skew)
  - Kurtosis: -0.3235 (Platykurtic / flatter than normal)

- **Visualizations Created:**
    - math_score_dist.png (Histogram with overlaid Normal distribution curve)
    - english_score_qq.png (Q-Q probability plot against theoretical normal quantiles)

Step 4: Linear Algebra Mini Task

- **Vectors (First 5 Students):**
    - $\mathbf{u}$ (Math Score) = $[44.4, 63.7, 88.7, 41.3, 86.9]$
    - $\mathbf{v}$ (Science Score) = $[69.3, 81.2, 76.6, 50.1, 89.5]$
- **Calculations:**
    - **Dot Product:** $\mathbf{u} \cdot \mathbf{v} = (44.4 \times 69.3) + \ldots + (86.9 \times 89.5) = 24{,}890.46$
    - **Norm-1 ($\|\mathbf{u}\|_1$):** $\sum|u_i| = 44.4 + 63.7 + 88.7 + 41.3 + 86.9 = 325.00$
    - **Norm-2 ($\|\mathbf{u}\|_2$):** $\sqrt{\sum u_i^2} = \sqrt{44.4^2 + \ldots + 86.9^2} = 152.16$
    - **Angle (θ):** $\arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}\right) = 11.08°$

HERE ARE KEY DATA-DRIVEN INSIGHTS DERIVED FROM THE STUDENT SCORES DATASET:

- **High Impact of Study Time:** Students who study more than **5 hours** have a **99.7%** chance of passing their courses.
- **Overall Passing Rate:** Across the entire student body, the baseline probability of passing is **60.6%**.
- **Study Duration Correlation:** There is a strong positive correlation (≈ 0.65) between the number of hours studied and whether a student ultimately passes or fails.
- **Math Score Parity:** Interestingly, the average Math Score for students who passed (**69.9**) is nearly identical to the average for students who failed (**70.2**), indicating that performance in individual subjects is secondary to overall study commitment or combined criteria.